Some reviews from vector spaces,

A vector spaces V over $\mathbb{C}$ is a linear space over $\mathbb{C}$,

also it is equipped with inner product: $V \times V \to \mathbb{R}$

1) $(X, Y) = \overline{(Y, X)}$.

2) $(\alpha X + \beta Y, Z) = \alpha (X, Z) + \beta (Y, Z)$

3). $(X, X) \geq 0$.

Then we have the norm $\|X\| := (X, X)^{\frac{1}{2}}$.

and it's easy to have: 1) If $(X, Y) = 0$ (which call $X \perp Y$).

then $\|X+Y\|^2 = \|X\|^2 + \|Y\|^2$.

2) $\forall X, Y \in V$. $|(X, Y)| \leq \|X\| \cdot \|Y\|$ (Cauchy-Schwarz inequality)

3). $\forall X, Y \in V$: $\|X+Y\| \leq \|X\| + \|Y\|$.

p.s. if V satisfies: 1) $\|X\| = 0 \Rightarrow X = 0$

2) V is complete, that is every Cauchy sequence converges

Thm: $\lim_{N \to \infty} \frac{1}{2\pi} \int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta = 0$.

pf: let $\mathcal{R} := \{ f \mid f \text{ is integrable on the circle} \}$.

$(f, g) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \overline{g(\theta)} \, d\theta$.

So the proposition equals to $\lim_{N \to \infty} \|f - S_N(f)\| = 0$

let $e_n(\theta) = e^{in\theta}$. then $\{e_n\}_{n \in \mathbb{Z}}$ is orthonormal.

$(f, e_n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta = \hat{f}(n) = a_n$.

Then $\left(f - \sum_{|n| \leq N} a_n e_n, \sum_{|n| \leq N} b_n e_n\right) = \sum_{|n| \leq N} b_n (f, e_n) - \left(\sum_{|n| \leq N} a_n e_n\right)\left(\sum_{|n| \leq N} b_n e_n\right)$

$= \sum_{|n| \leq N} a_n b_n - \sum_{|n| \leq N} a_n b_n = 0$

So $\left(f - \sum_{|n| \leq N} a_n e_n\right) \perp \sum_{|n| \leq N} b_n e_n$ $\left(\sum_{|n| \leq N} a_n e_n = S_N(f).\right)$

Thus: $\|f\|^2 = \|f - \sum_{|n| \leq N} a_n e_n\|^2 + \|\sum_{|n| \leq N} a_n e_n\|^2 = \|f - \sum_{|n| \leq N} a_n e_n\|^2 + \sum_{|n| \leq N} |a_n|^2$.

Lemma: $\|f - S_N(f)\| \leq \|f - \sum_{|n| \leq N} c_n e_n\|$.

pf: $\|f - \sum_{|n| \leq N} c_n e_n\| = \|f - S_N(f) + \sum_{|n| \leq N} b_n e_n\|$ $b_n = a_n - c_n$

$(f - S_N(f)) \perp \sum_{|n| \leq N} b_n e_n \Rightarrow \|f - \sum_{|n| \leq N} c_n e_n\| = \sqrt{\|f - S_N(f)\|^2 + \|\sum_{|n| \leq N} b_n e_n\|^2} \geq \|f - S_N(f)\|$

Back to the proof,

1. f is continuous, then $\forall \varepsilon > 0 \quad \exists P$ (三角多项式) $\deg P = M$

   s.t. $|f(\theta) - P(\theta)| < \varepsilon \quad \forall \theta \in [-\pi, \pi]$

   Then: by integration, $\|f - P\| < \varepsilon$.

   with the previous lemma: $\|f - S_N(f)\| \le \|f - P\| < \varepsilon$. $\forall N \ge M$

2. Lemma 2. f is integrable on the circle. $|f| \le B$

   Then there exists $\{f_k\}_{k=1}^{\infty}$ f_k is continuous

   s.t. $\sup_{x \in [-\pi, \pi]} |f_k(x)| \le B$.

   $\int_{-\pi}^{\pi} |f(x) - f_k(x)| dx \to 0 \quad (k \to \infty)$ proof is left.

$\forall \varepsilon > 0$.

3. Back to the proof. $\forall$ choose g. (continuous)

   satisfies. $\sup |g(\theta)| \le \sup |f(\theta)| = B$.

   $\int_0^{2\pi} |f(\theta) - g(\theta)| d\theta < \varepsilon^2$.

   $\|f - g\|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta) - g(\theta)|^2 d\theta \le 2B\varepsilon^2$.

   then choose P. s.t $\|g - P\| < \varepsilon$. $\deg P = M$

   $\|P - S_N(f)\| < \varepsilon$. $\forall N \ge M$

   then $\|f - S_N(f)\| \le \|f - g\| + \|g - P\| + \|P - S_N(f)\| < C\varepsilon$. C is a $\forall$ constant

That is $\lim_{N \to \infty} \|f - S_N(f)\| = 0$ □

1) $\sum_{n=-\infty}^{\infty} |a_n|^2$ converges. then $\|f\|^2 = \sum_{n=-\infty}^{\infty} |a_n|^2$. a_n is fourier coeffiencient

that is $\frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta = \sum_{n=-\infty}^{\infty} |a_n|^2$. (Parseval's identity)

2). If $\{e_n\}$ is other orthonormal family of $\mathcal{R}$. let $a_n = (f, e_n)$

then we have: $\sum_{n=-\infty}^{\infty} |a_n|^2 \le \|f\|^2$. (Bessel inequality)

Thm. If $f \in R(\mathbb{R}/2\pi\mathbb{Z}) \cap \{\theta_0\}$. then. $\lim_{N\to\infty} S_N(f)(\theta_0) = \theta_0$

pf: $f \in C^1(\mathbb{R}/2\pi\mathbb{Z}) \cap \{\theta_0\}$. then: $\lim_{\theta\to\theta_0} \frac{f(\theta)-f(\theta_0)}{\theta-\theta_0}$ exists, $\triangleq f'(\theta_0)$

Defi $F(t) = \begin{cases} \frac{f(\theta_0-t)-f(\theta_0)}{t} & |t|\neq 0. \quad |t|<\pi. \\ -f'(\theta_0). & t=0. \end{cases}$

$F(t) \in C^0((\mathbb{R}/2\pi\mathbb{Z})^\bullet)$

then we have F is integrable on the circle.

So. $S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int_0^{2\pi} F(t)\, t D_N(t)\, dt.$

Since. $D_N(t) = \frac{\sin(N+\frac{1}{2})t}{\sin\frac{1}{2}t}.$

then $tD_N(t) = \frac{t\sin(N+\frac{1}{2})t}{\sin\frac{1}{2}t}$ is bounded.

$\frac{t}{\sin\frac{1}{2}t} \in C([-\pi,\pi])$.

$\sin(N+\frac{1}{2})t = \sin(Nt)\cos\frac{1}{2}t + \sin\frac{1}{2}t\cos(Nt). \Rightarrow$

$\begin{cases} \int_0^{2\pi} F(t)\sin(Nt)dt \to 0 & (N\to\infty) \\ \int_0^{2\pi} F(t)\cos(Nt)dt \to 0 & (N\to\infty) \end{cases}$ (Riemann - Lebesgue Lemma)

then $\lim_{N\to\infty} S_N f(\theta_0) = f(\theta_0)$ □.

p.s. However, when f simply $\in C^0(\mathbb{R}\setminus 2\pi\mathbb{Z})$., then it doesn't always satisfies:

$\lim_{N\to\infty} S_N(f) = f$

Consider the example: $\sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{n}$.

We can prove that it doesn't converges to any integrable f !!!

□

fourier transform on $\mathbb{R}$.

1. Def: ① consider $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-2\pi i n x}\, dx$, $f(x) \sim \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inx} dx$

its continuous version $\hat{f}(\xi) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x\xi}\, dx$.

then $f(x) \sim \int_{-\infty}^{+\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi$.

② $S(\mathbb{R})$. (Schwarz space): if f satisfies $f \leq \frac{A'}{1+x^2}$ (A' is a dim=1's. constant). , then

$S(\mathbb{R}) := \{f \mid f^{(k)}$ exists. for all $k \geq 0$, $\wedge$. $\sup_{x\in\mathbb{R}} |x|^k |f^{(\alpha)}(x)| < \infty$ for all $k, \alpha \geq 0$

2. fourier transform on $S(\mathbb{R})$:

then we have.

1). $\hat{f}(\xi) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x\xi}\, dx = \int_{-\infty}^{+\infty} f(x+h) e^{-2\pi i x\xi}\, dx$

2) $\hat{f}(\xi+h) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x\xi} \cdot e^{-2\pi i x h}\, dx = \hat{g}(\xi)$ $g = f \cdot e^{-2\pi i x h}$.

3). proof: $f(\delta x) \to \delta^{-1} \hat{f}(\delta^{-1}\xi)$.

pf: let $g(x) = f(\delta x)$. $\hat{g}(\xi) = \int_{-\infty}^{+\infty} f(\delta x) e^{-2\pi i x\xi}\, dx$.

$$\delta^{-1}\hat{f}(\delta^{-1}\xi) = \delta^{-1}\int_{-\infty}^{+\infty} f(x) e^{-2\pi i x \delta^{-1}\xi}\, dx$$

$$\overset{x=\delta y}{=\!=} \int_{-\infty}^{+\infty} f(\delta y) e^{-2\pi i \xi y}\, dy = \hat{g}(\xi).$$

4). $2\pi i\xi \hat{f}(\xi) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x\xi}\, dx \cdot (2\pi i\xi) = -\int_{-\infty}^{+\infty} f(x)\, d\left(e^{-2\pi i x\xi}\right)$.

$$= -f(x) e^{-2\pi i x\xi}\Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} f'(x) e^{-2\pi i x\xi}\, dx.$$

$$= \hat{f'}(\xi)$$

5). $\frac{d}{d\xi}\hat{f}(\xi) = \frac{d}{d\xi}\left(\int_{-\infty}^{+\infty} f(x) e^{-2\pi i x\xi}\, dx\right) = \lim_{h\to 0} \frac{\hat{f}(\xi+h) - \hat{f}(\xi)}{h}$.

$$= \lim_{h\to 0} \int_{-\infty}^{+\infty} f(x) \frac{e^{-2\pi i x(\xi+h)} - e^{-2\pi i x\xi}}{h}$$

$$= \lim_{h\to 0} \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x\xi} \left(\frac{e^{-2\pi i x h} - 1}{h}\right) dx.$$

不太严格 $\int_{-\infty}^{+\infty} f(x) e^{-2\pi i x\xi} (-2\pi i x)\, dx$

用含 ε-δ.

$$= \widehat{-2\pi i x f}(\xi). \quad \square.$$

we can conclade that: $f(x) \to \hat{f}(\xi)$ means $\hat{f}$ is the fourier transform of f

(i) $f(x+h) \to \hat{f}(\xi)\, e^{2\pi i h\xi}$ $\forall h \in \mathbb{R}$

(ii) $f(x)e^{-2\pi i x h} \to \hat{f}(\xi+h)$ $\forall h \in \mathbb{R}$.

(iii) $f(\delta x) \to \delta^{-1}\hat{f}(\delta^{-1}\xi)$ $\forall \delta > 0$.

(iv) $f'(x) \to 2\pi i \xi \hat{f}(\xi)$

(v) $-2\pi i x f(x) \to \frac{d}{d\xi}\hat{f}(\xi)$.

Thm: If $f \in S(\mathbb{R})$ then $\hat{f} \in S(\mathbb{R})$.

Pf: $(\xi^k)\left(\frac{d}{d\xi}\right)^l \hat{f}(\xi) \leftarrow \frac{1}{(2\pi i)^k}\left(\frac{d}{d\xi}\right)^k [(-2\pi i x)^l f(x)]$ (based on (iv) & (v)) $\to$ it $\in S(\mathbb{R})$.

and we have if $f \in S(\mathbb{R})$ then $\hat{f}$ is bouned

so. $(\xi^k)\left(\frac{d}{d\xi}\right)^l \hat{f}(\xi)$ is bounded. then $\hat{f} \in S(\mathbb{R})$ □.

Good Kernels: let: $K_\delta(x) = \delta^{-\frac{1}{2}} e^{\frac{-\pi x^2}{\delta}}$ then we have (i) $\int_{-\infty}^{+\infty} K_\delta(x)\,dx = 1$

(ii). $\int_{-\infty}^{+\infty} |K_\delta(x)|\,dx \le M$

(iii) For $\forall \eta > 0$. we have $\int_{|x|>\eta} |K_\delta(x)|\,dx \to 0$ as $\delta \to 0$.

Pf: (i). $\int_{-\infty}^{+\infty} \delta^{-\frac{1}{2}} e^{\frac{-\pi x^2}{\delta}}\,dx = \int_{-\infty}^{+\infty} e^{-\pi(x\cdot\delta^{-\frac{1}{2}})^2} d(x\cdot\delta^{-\frac{1}{2}}) = \int_{-\infty}^{+\infty} e^{-\pi t^2}dt = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} e^{-\pi(x^2+y^2)}dx\,dy$

$\int_0^{2\pi}\int_0^{+\infty} e^{-\pi r^2} r\,dr\,d\theta$

$= \int_0^{+\infty} 2\pi r e^{-\pi r^2}dr = -e^{-\pi r^2}\Big|_0^{+\infty}$

$= 1$

(ii). $K_\delta(x) \ge 0 \Rightarrow \int_{-\infty}^{+\infty} |K_\delta(x)|\,dx = 1 \le M$

(iii). $\int_{|x|>\eta} |K_\delta(x)|\,dx \overset{t}{=} \int_{|y|>\eta\cdot\delta^{-\frac{1}{2}}} e^{-\pi y^2}dy \to 0$ $(\delta \to 0)$ □.

2 Convolution on $\mathbb{R}$:

Definition: $(f*g)(x) = \int_{-\infty}^{+\infty} f(x-t)\, g(t)\, dt$, $(f, g \in S(\mathbb{R}))$

Thm: If $f \in S(\mathbb{R})$ then $(f * K_\delta)(x) \to f(x)$ uniformly in x as $\delta \to 0$

pf: $|(f*K_\delta)(x) - f(x)| = \left| \int_{-\infty}^{+\infty} K_\delta(t)\,(f(x-t) - f(x))\, dt \right|$

$$\leq \int_{-\infty}^{+\infty} |K_\delta(t)|\, |f(x-t) - f(x)|\, dt$$

$$= \left(\int_{|t|>\eta} + \int_{|t|<\eta}\right)\left(|K_\delta(t)|\, |f(x-t) - f(x)|\, dt\right).$$

1) $\forall \varepsilon > 0$. $\exists R > 0$ s.t. $|f(x)| < \frac{\varepsilon}{4}$ when $|x| \geq R$ Also $f \in C([R, R])$

So it's easy to conclude that f is uniformly continuous on $\mathbb{R}$

Also f is bounded.

So $\int_{|t|>\eta} |K_\delta(t)|\, |f(x-t) - f(x)|\, dt \leq \varepsilon \cdot 2 \sup_{x \in \mathbb{R}} |f(x)|$ $(\delta \to 0)$

$\int_{|t|<\eta} |K_\delta(t)|\, |f(x-t) - f(x)|\, dt \leq \varepsilon \cdot M$ $(|t| < \eta)$

$\Rightarrow |(f * K_\delta)(x) - f(x)| \to 0$ uniformly in x. as $\delta \to 0$ □.

Thm: If $f \in S(\mathbb{R})$ then $f(x) = \int_{-\infty}^{+\infty} \hat{f}(\xi)\, e^{2\pi i x \xi}\, d\xi$.

pf:

Lemma: If $f, g \in S(\mathbb{R})$. then $\int_{-\infty}^{+\infty} f(x)\, \hat{g}(x)\, dx = \int_{-\infty}^{+\infty} \hat{f}(y)\, g(y)\, dy$.

pf: note that $F(x,y) = f(x)\, g(y)\, e^{-2\pi i x y}$

we have $|F(x,y)| < \frac{A}{(1+x^2)(1+y^2)}$ A is a constant.

Then we have $F_1(x) = \int_{-\infty}^{+\infty} F(x,y)\, dy \in M(\mathbb{R})$, $F_2(y) = \int_{-\infty}^{+\infty} F(x,y)\, dx \in M(\mathbb{R})$

Also $\int_{-\infty}^{+\infty} F_1(x)\, dx = \int_{-\infty}^{+\infty} F_2(y)\, dy$.

$\Rightarrow \int_{-\infty}^{+\infty} f(x)\, \hat{g}(x)\, dx = \int_{-\infty}^{+\infty} \hat{f}(y)\, g(y)\, dy$ □.

Back to the proof:

let $G_\delta(x) = e^{-\pi \delta x^2}$ s.t. $\hat{G}_\delta(\xi) = \int_{-\infty}^{+\infty} G_\delta(x)\, e^{-2\pi i x \xi}\, dx$

$$= \int_{-\infty}^{+\infty} e^{-\pi \delta x^2} \cdot e^{-2\pi i x \xi}\, dx$$

$\Rightarrow \int_{-\infty}^{+\infty} f(x) K_\delta(x)\, dx = \int_{-\infty}^{+\infty} \hat{f}(\xi) G_\delta(\xi)\, d\xi \equiv K_\delta(\xi)$

let: $\delta \to 0$ we have $f(0) = \int_{-\infty}^{+\infty} \hat{f}(\xi)\, d\xi$

Then let $F(y) = f(y+x)$.

s.t. $f(x) = F(0) = \int_{-\infty}^{+\infty} \hat{F}(\xi)\, d\xi = \int_{-\infty}^{+\infty} \hat{f}(\xi)\, e^{2\pi i x \xi}\, d\xi$ □.

Corollary: define two mappings,

1) $\mathcal{F}: S(\mathbb{R}) \to S(\mathbb{R})$ $\quad f(\xi) \mapsto \int_{-\infty}^{+\infty} f(x) e^{-\pi i x \xi} dx$ (fourier transform)

2) $\mathcal{F}^*: S(\mathbb{R}) \to S(\mathbb{R})$ $\quad g(x) \mapsto \int_{-\infty}^{+\infty} g(\xi) e^{2\pi i x \xi} d\xi$

then we have $\mathcal{F}^* \circ \mathcal{F} = \mathcal{F} \circ \mathcal{F}^* = I$

pf: $\mathcal{F}^* \circ \mathcal{F}(f)(\xi) = \mathcal{F}^*\left(\int_{-\infty}^{+\infty} f(x) e^{-\pi i x \xi} dx\right) = \mathcal{F}^*(\hat{f})(\xi) = \int_{-\infty}^{+\infty} \hat{f}(x) e^{2\pi i x \xi} dx$

$= f(\xi)$

$\Rightarrow \mathcal{F}^* \circ \mathcal{F} = I$

since $\mathcal{F}^*(f)(y) = \mathcal{F}(f)(-y)$ so. $\mathcal{F} \circ \mathcal{F}^*(f)(\xi) = \mathcal{F}(\mathcal{F}^*(f)(-\xi))$

$= \mathcal{F}^*(\mathcal{F}(f)(\xi)) = f(\xi)$

$\therefore$ thus $\mathcal{F}$ is a bijective mapping on $S(\mathbb{R})$ $\square$.

Convolutions on $S(\mathbb{R})$. If $f, g \in S(\mathbb{R})$

(i) $f * g \in S(\mathbb{R})$

(ii). $f * g = g * f$.

(iii). $\widehat{(f*g)}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

pf: (i). $\sup_x |x^l (f*g)(x)| = \sup_x |x^l \int_{-\infty}^{+\infty} f(x-t) g(t) dt|$

$\leq A_l \int_{-\infty}^{+\infty} |f(y)| (1+|y|)^l dy < \infty$.

$\left(\frac{d}{dx}\right)^k (f*g)(x) = \left(f * \left(\frac{d}{dx}\right)^k g\right)(x)$

$\left(\frac{d}{dx}\int_{-\infty}^{+\infty} f(x-t) g(t) dt = \int_{-\infty}^{+\infty} \frac{d}{dx} g(x-t) f(t) dt\right.$ (for the uniform convergence).)

thus. $\sup \left(\frac{d}{dx}\right)^k (f*g)(x) < \infty$.

(ii) trival.

(iii). let $F(x,y) = f(y) g(x-y) e^{-2\pi i x \xi}$.

$F_1(x) = \int_{-\infty}^{+\infty} F(x,y) dy = (f*g)(x) e^{-2\pi i x \xi}$.

$F_2(y) = \int_{-\infty}^{+\infty} F(x,y) dx = f(y) e^{-2\pi i y \xi} \hat{g}(\xi)$

And. $\int_{-\infty}^{+\infty} F_1(x) dx = \int_{-\infty}^{+\infty} F_2(y) dy$ that is $\widehat{f*g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$ $\square$.

Thm: (Plancherel). If $f \in S(\mathbb{R})$. then $\|\hat{f}\| = \|f\|$.

proof: $\|f\| = \left(\int_{-\infty}^{+\infty} |f(x)|^2 dx\right)^{\frac{1}{2}}$

let $f^b(x) = \overline{f(-x)}$. $\quad \hat{f^b}(\xi) = \int_{-\infty}^{+\infty} \overline{f(-x)} e^{-2\pi i x \xi} dx$

$= \int_{-\infty}^{+\infty} \overline{f(-x) e^{-2\pi i (-x)\xi}} dx = \overline{\hat{f}(\xi)}$.

let $h = f * f^b = \int_{-\infty}^{+\infty} f(x-t) f^b(t) dt$

the $\hat{h}(\xi) = |\hat{f}(\xi)|^2$ $\quad h(0) = \left(\int_{-\infty}^{+\infty} |f(x)|^2 dx\right)^2$

since $h(0) = \int_{-\infty}^{+\infty} \hat{h}(\xi) d\xi$. $\Rightarrow \|f\| = \|\hat{f}\|$ $\square$.

Thm: (Poisson summation formula) If $f \in S(\mathbb{R})$ then $\underbrace{\sum_{n=-\infty}^{\infty} f(x+n)}_{G} = \underbrace{\sum_{n=-\infty}^{\infty} \hat{f}(n)\, e^{2\pi i n x}}_{H}$

pf: for the uniform and absolute convergence of both two sides (G & H are continuous)

We just need to prove the left fourier coefficients are equal to the right fourier coefficients.

the mth fourier coefficient: $\int_0^1 \left(\sum_{n=-\infty}^{\infty} f(x+n)\right) e^{-2\pi i m x}\, dx = \sum_{n=-\infty}^{\infty} \int_0^1 f(x+n)\, e^{-2\pi i m x}\, dx$

$$= \sum_{n=-\infty}^{\infty} \int_n^{n+1} f(y)\, e^{-2\pi i y m}\, dy$$

$$= \int_{-\infty}^{+\infty} f(y)\, e^{-2\pi i m y}\, dy$$

$$= \hat{f}(m)$$

$\Rightarrow \sum_{n=-\infty}^{\infty} f(x+n) = \sum_{n=-\infty}^{\infty} \hat{f}(n)\, e^{2\pi i n x}$. □